



PetPulse

Smart Care for Every Paw

PRESENTED TO
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PROGRESS & TECHNICAL VISION SHOWCASE

Introduction & Problem Statement

Why PetPulse exists and the core challenges we are solving.

Introduction

PetPulse is a unified, web-based platform designed to seamlessly connect pet owners with veterinarians and essential pet service providers.

By integrating robust Artificial Intelligence and high-end security mechanisms, we ensure a scalable, reliable, and secure environment for all users.

The Problem

- **Scattered Services:** Users struggle to find reliable help quickly across fragmented platforms.
- **Emergency Delays:** Lack of intelligent support during urgent, life-threatening situations.
- **Security Risks:** Existing platforms lack strong data protection, reducing user trust.
- **Inefficiency:** Manual booking and matching processes are slow and outdated.

Project Objectives

Our vision for a centralized, secure, and intelligent ecosystem.



Centralized Platform

Develop a single, unified pet-care hub.



Improved Accessibility

Frictionless access to veterinary services.



AI-Based Assistance

Smart triage and booking automation.



Data Security

Ensure absolute privacy and system integrity.



Scalability & Reliability

Build a practical system ready for real-world application.

Core Web Features

Hover over our core capabilities to view their functionality.



Vet Booking

Find nearby verified vets, check ratings, and book appointments instantly.

Value: Trusted pet care access.



Lost & Found

Post alerts with precise GPS locations to immediately reach nearby users.

Value: Community-driven recovery.



Adoption Services

Browse pet profiles for adoption or mating and contact owners securely.

Value: Safe, responsible connections.



Real-Time Chat

WebSocket-powered chat. Connect directly with vets, trainers, and owners.

Value: Active pet community.



Training & Behavior

Discover certified trainers and structured, professional guidance plans.

Value: Improves daily pet care.



Service Matching

AI-driven location and need-based matching for specialized pet services.

Value: Personalized experience.



RBAC System

Role-Based Access Control for Users, Veterinarians, and Administrators.

Value: Structured permission layers.



Admin Dashboard

Comprehensive oversight for moderation, analytics, and platform health.

Value: Ensures platform trust.

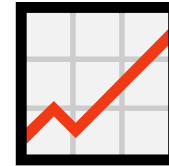
[Click for Details]

Admin Dashboard

A comprehensive command center for platform management and data analytics.

Key Capabilities

- **Real-Time Analytics:** Monitor active users, system health, and booking statistics.
- **User Moderation:** Approve or suspend accounts, ensuring a safe community.
- **Financial Oversight:** Track payment flows and subscription statuses.
- **Content Management:** Moderate lost & found alerts and adoption posts.



Data-Driven Insights

Empowering our admins with actionable intelligence to maintain platform integrity.

AI Integration: Overview

Enhancing functionality and supporting intelligent decision-making.



Artificial Intelligence

Our AI engine operates as the core brain of PetPulse, eliminating friction and accelerating response times during critical moments.

Symptom Analysis & Smart Triage

- **Natural Language Processing:** Understands user descriptions of pet ailments in real-time.
- **Urgency Classification:** Automatically flags cases as *Routine*, *Urgent*, or *Emergency*.
- **Pre-Diagnosis Context:** Provides veterinarians with structured preliminary data before the visit.

```
> INPUT: "Cat is lethargic and vomiting yellow liquid"
```

Agentic AGI Engine

Watch our autonomous AI process natural language and execute complex backend workflows without human intervention.

petpulse-agentic-microservice — bash

```
> Waiting for incoming socket connection...  
  
> USER PROMPT: "My dog Max is limping slightly on his front paw. Can I see Dr. Smith tomorrow afternoon?"  
  
> [AGI] Analyzing intent & triage risk level...  
  
> [AGI] Risk Level: NORMAL. Querying PostGIS for Dr. Smith... [FOUND]  
  
> [AGI] Checking calendar availability for tomorrow afternoon... [14:00 AVAILABLE]
```

```
{  
  "action": "CREATE_BOOKING",  
  "confidence": 0.98,  
  "payload": {  
    "vet_id": "v_74829",  
    "pet_id": "p_max_01",  
    "time": "2026-05-06T14:00:00Z",  
    "notes": "Slight limp on front paw. Non-urgent."  
  }  
}
```

Intelligent Matching System

Hyper-personalized service recommendations driven by data.

How It Works

1. **Profile Analysis:** Examines pet breed, age, and medical history.
2. **Geospatial Querying:** Utilizes PostGIS to find nearest available providers within a defined radius.
3. **Requirement Matching:** Aligns user needs with provider skill sets.
4. **Dynamic Ranking:** Sorts results based on real-time availability and relevance.

Efficiency Gain

Manual Booking Process ~5 Minutes



PetPulse Agentic AGI 2.3 Seconds



Interactive System Architecture

Simulate a real-time data flow through our zero-trust pipeline.



Client App



API Gateway
JWT / RBAC



Agentic AGI



PostgreSQL
+ PostGIS

Threat Protection & Monitoring

Active defense mechanisms against common attack vectors.

`{{ }}` View Source Logic

Active Defenses

- **SQL Injection Prevention:** Strict use of parameterized queries and ORM layers.
- **XSS Protection:** Aggressive input sanitization and secure Content Security Policies (CSP).
- **Rate Limiting:** Preventing DDoS and brute-force authentication attacks.

LIVE TRAFFIC LOG

```
[OK] GET
/api/vets/123 -
200ms

[OK] POST
/api/auth/login -
150ms

[WARN] DROP TABLE
detected from IP
192.168.1.45

[BLOCKED] SQLi
payload
neutralized. IP
Banned.

[OK] WSS /chat
connected
```



System Integrity

Logging,
Monitoring & Alert
Mechanisms Active.

Methodology & Future Work

Our development lifecycle and the roadmap ahead.

Development Workflow

Jira Task Tracking

Frontend UI Complete

PostgreSQL DB Ready

APIs Connected

- **Jira Integration:** Enabled us to track sprints and organize our tasks efficiently across all 3 teams.
- **Full Stack Connected:** The frontend user screens are fully integrated with our robust backend PostgreSQL database.
- **Systematic Execution:** Ensuring efficiency, organization, and a high-quality product ready for deployment.

Future Roadmap

- **Mobile App:** Native Android & iOS development.
- **Enhanced AI:** Larger dataset training for higher accuracy.
- **IoT Integration:** Smart collars for real-time proactive health monitoring.
- **Cloud Expansion:** Elastic scalability.

What We Have Accomplished So Far

[Click the items below to actively check off our progress.](#)

-  **Frontend Foundation:** Fully responsive user screens complete.
-  **Database Architecture:** PostgreSQL & PostGIS schemas finalized.
-  **Full-Stack Integration:** APIs actively connecting client to server.
-  **Security Implementation:** JWT Auth and Role-Based Access active.
-  **AI Integration:** Currently training and deploying AGI models.

Conclusion

PetPulse combines web technologies, Artificial Intelligence, and cybersecurity practices to deliver a comprehensive, impactful solution for the modern pet-care industry.

Thank You.

Any Questions?